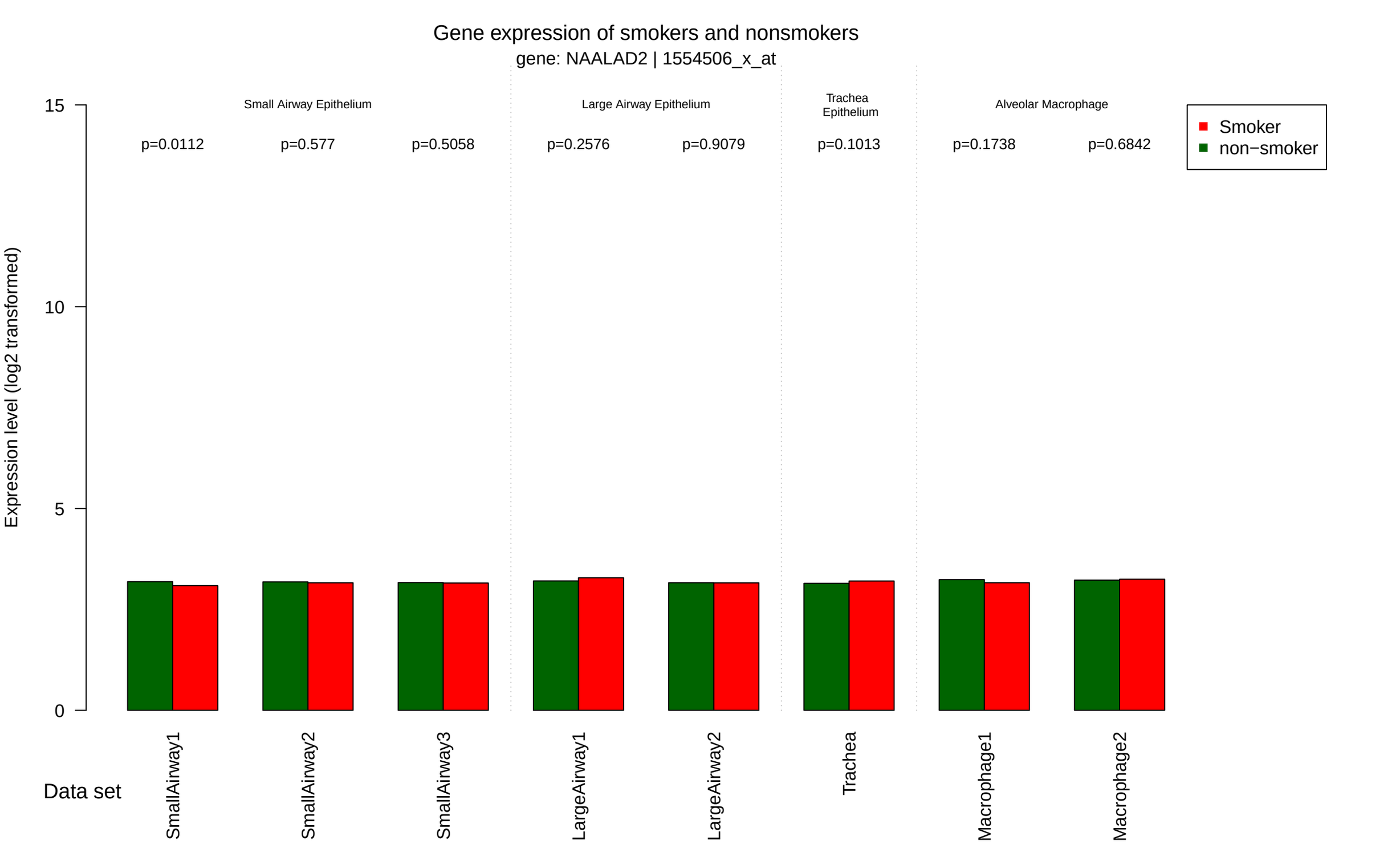
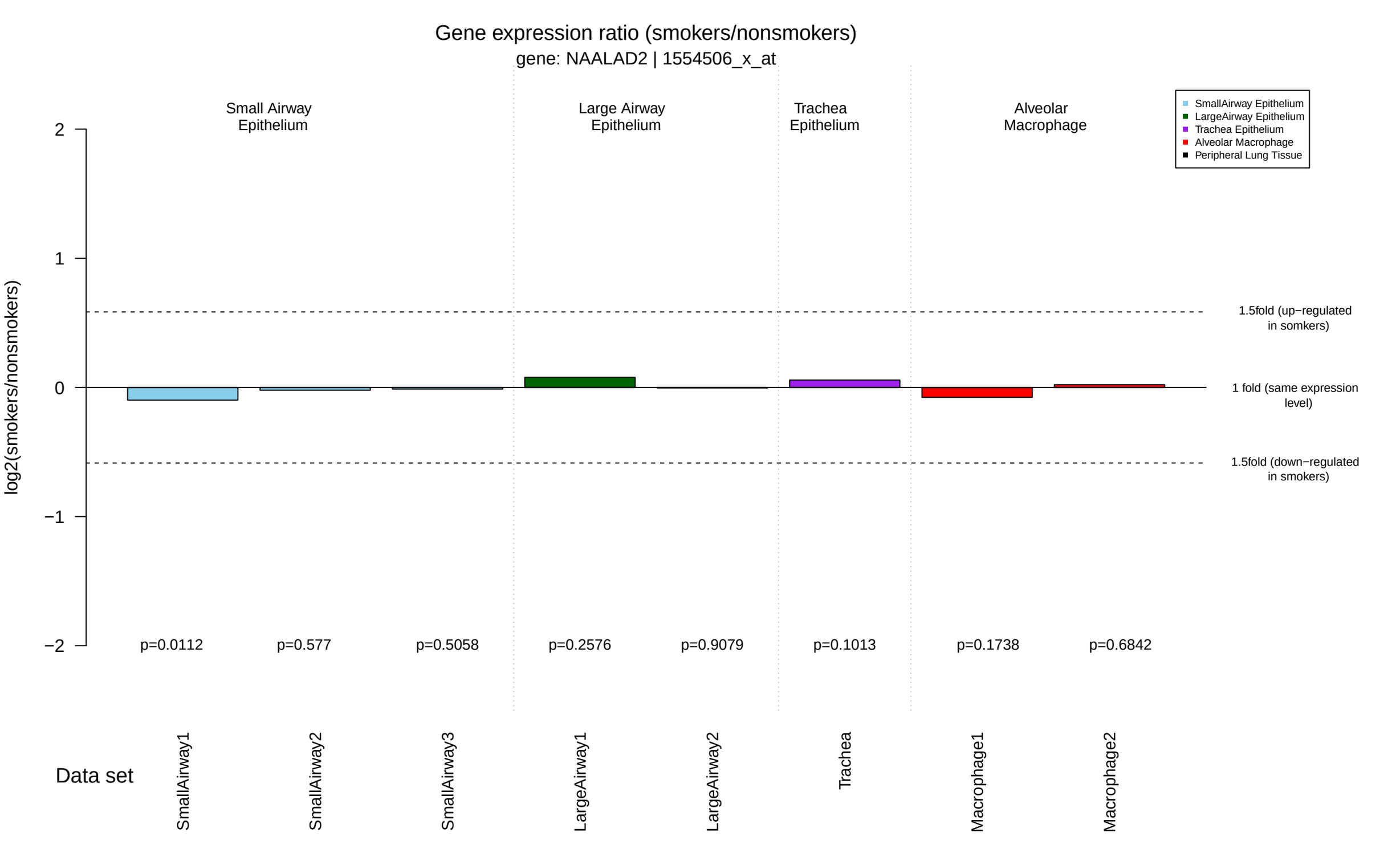
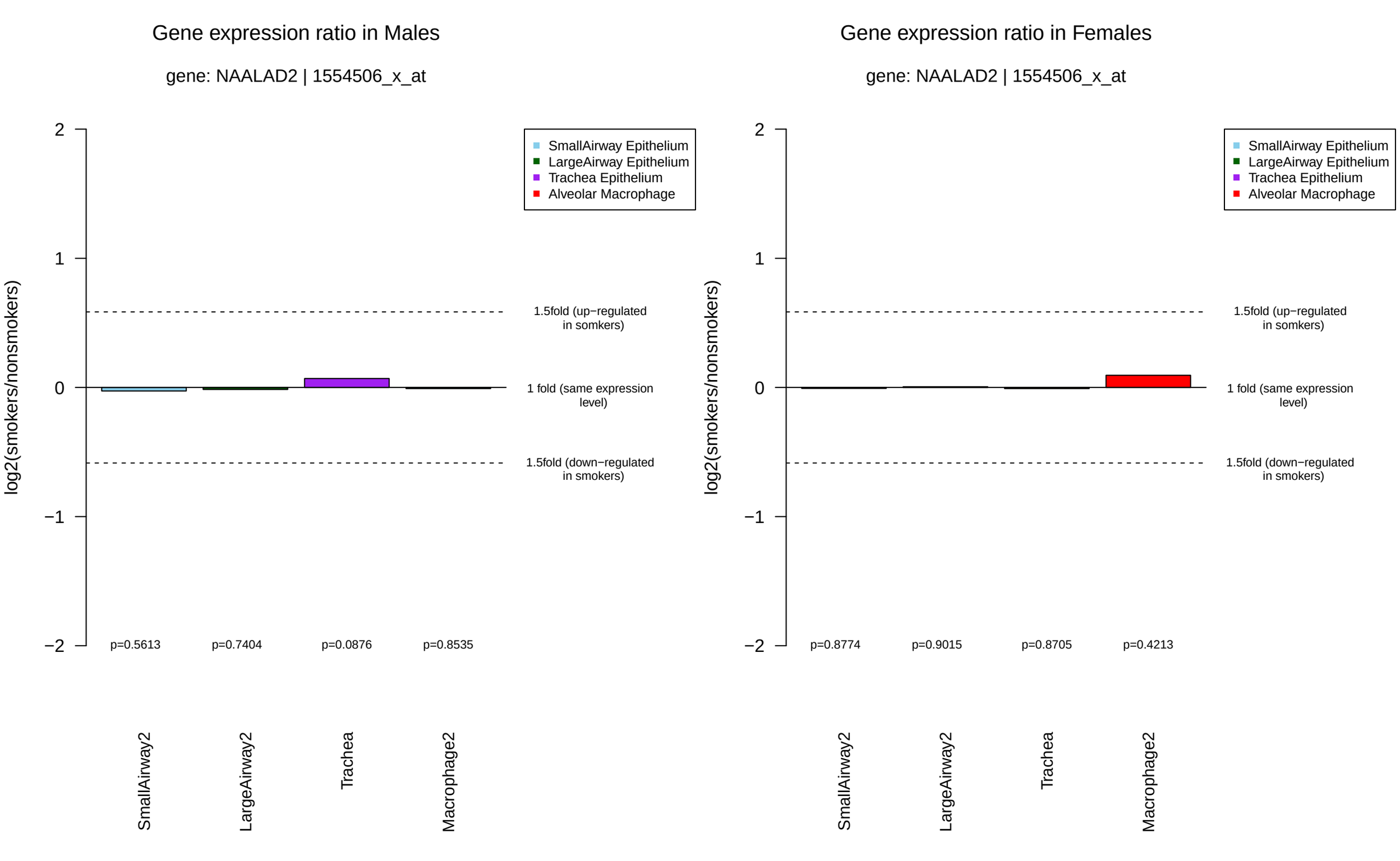
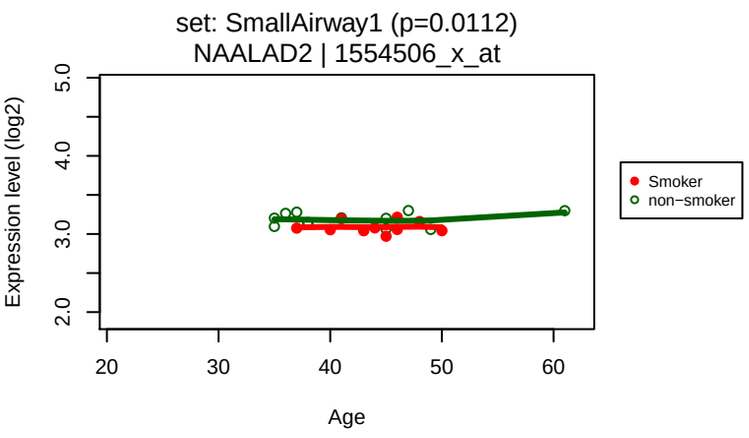
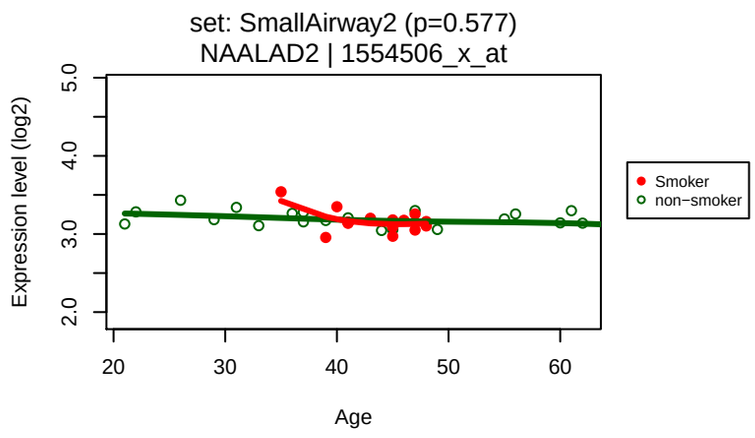
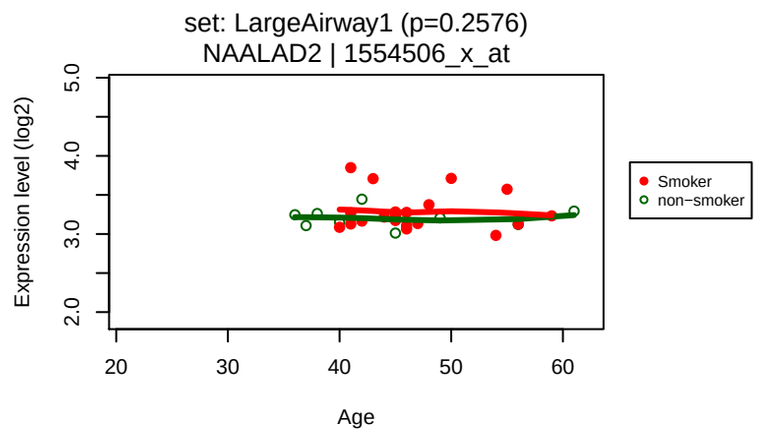

Gene expression of smokers and nonsmokers
gene: NAALAD2 | 1554506_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0112
p=0.577
p=0.5058
p=0.2576
p=0.9079
p=0.1013
p=0.1738
p=0.6842
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

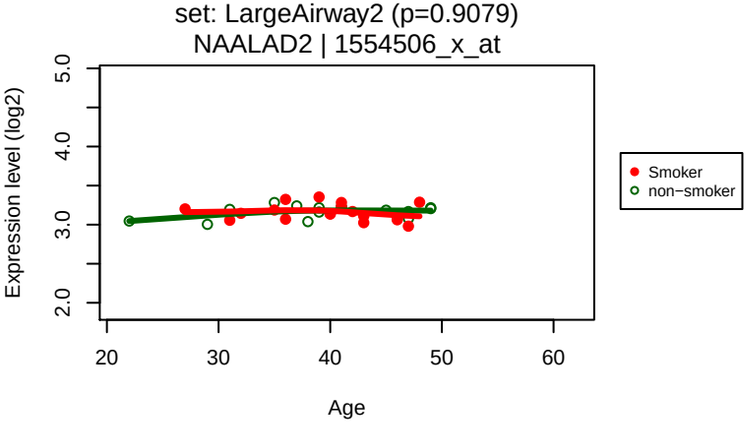

Gene expression ratio (smokers/nonsmokers)
gene: NAALAD2 | 1554506_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0112
p=0.577
p=0.5058
p=0.2576
p=0.9079
p=0.1013
p=0.1738
p=0.6842
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

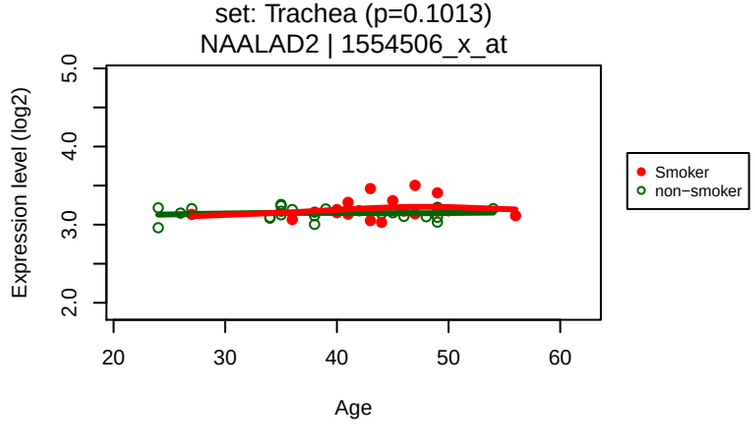

Gene expression ratio in Males
gene: NAALAD2 | 1554506_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5613
p=0.7404
p=0.0876
p=0.8535
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NAALAD2 | 1554506_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8774
p=0.9015
p=0.8705
p=0.4213
SmallAirway2
LargeAirway2
Trachea
Macrophage2

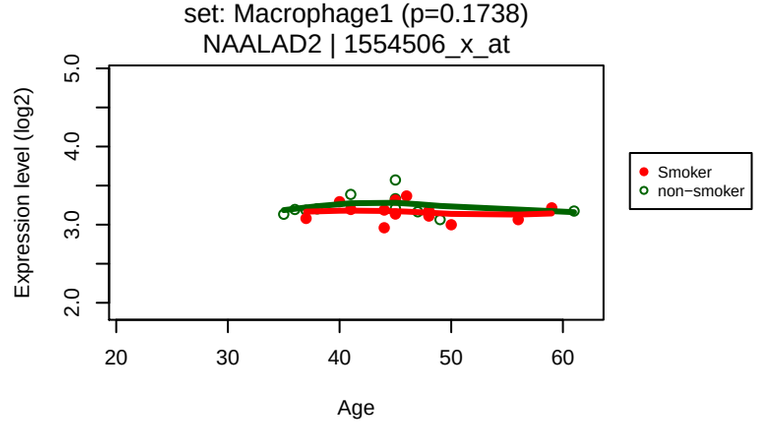

set: SmallAirway1 (p=0.0112)
NAALAD2 | 1554506_x_at
Expression level (log2)
Age
Smoker
non−smoker

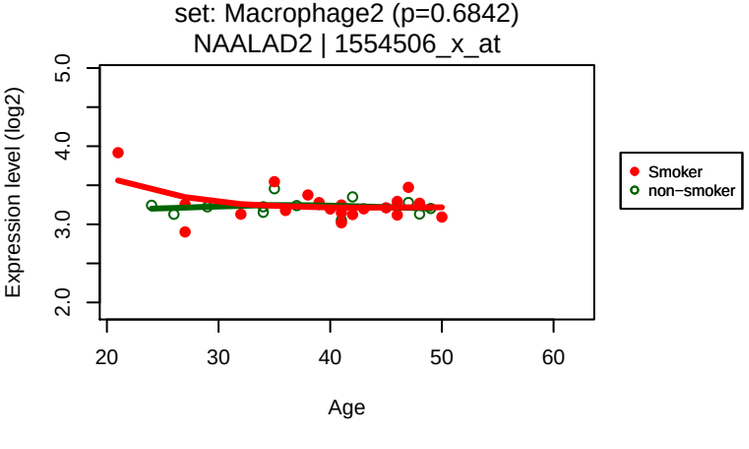

set: SmallAirway2 (p=0.577)
NAALAD2 | 1554506_x_at
Expression level (log2)
Age
Smoker
non−smoker

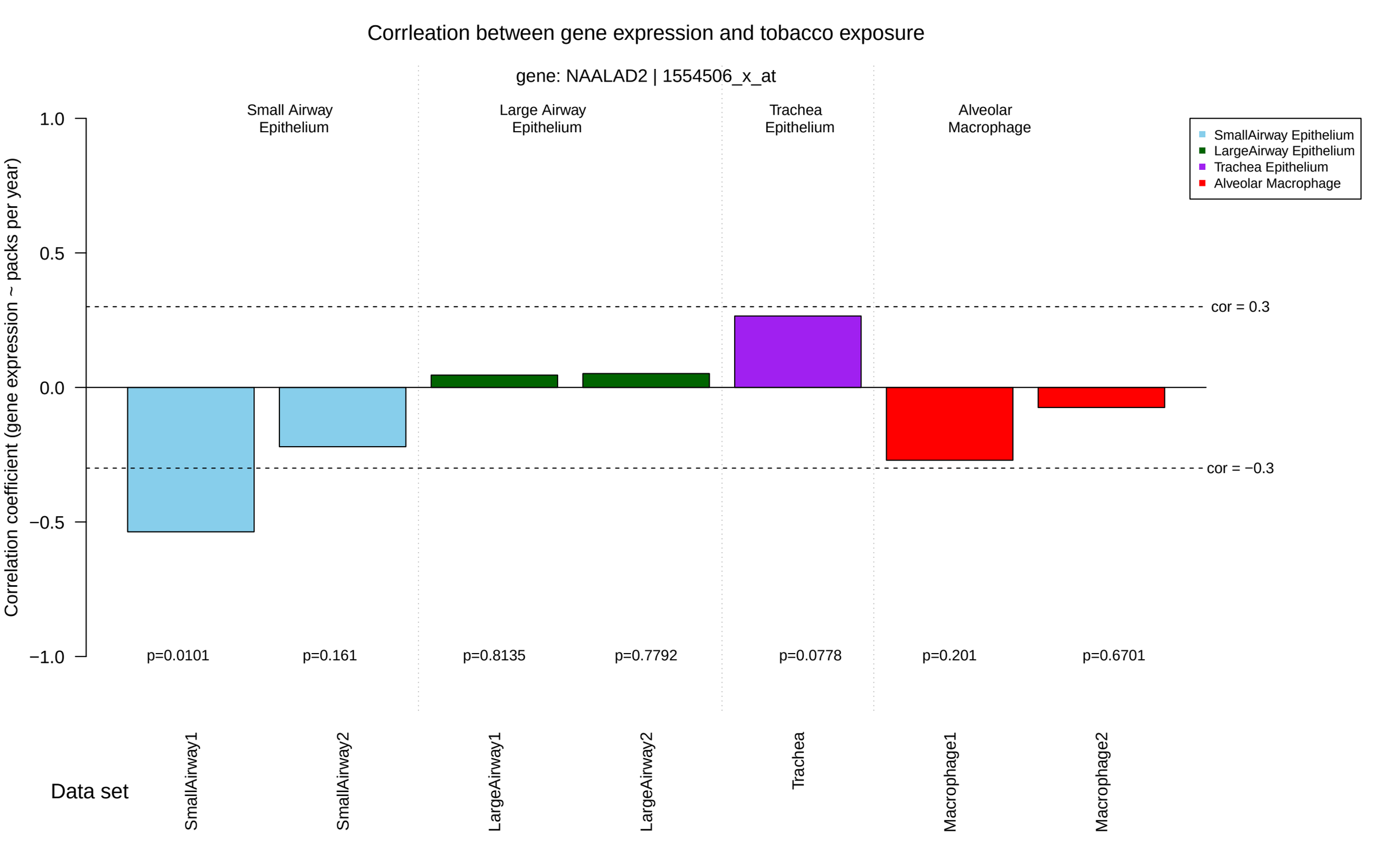

set: LargeAirway1 (p=0.2576)
NAALAD2 | 1554506_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9079)
NAALAD2 | 1554506_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1013)
NAALAD2 | 1554506_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1738)
NAALAD2 | 1554506_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6842)
NAALAD2 | 1554506_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NAALAD2 | 1554506_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0101
p=0.161
p=0.8135
p=0.7792
p=0.0778
p=0.201
p=0.6701
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2